Author Response to Reviewer u34v (Rating: 4, Borderline Accept)

Anonymous Authors

We thank Reviewer u34v for the detailed and constructive suggestions on presentation and clarity. Below we address each point in detail. All changes in the revised manuscript are marked in blue text; line references (e.g., `[L42--44]`) refer to the revised `main.tex`.

R3.1: Manuscript feels too fragmented due to titled paragraphs. **Response:** We have revised the Discussion (Section 6, `[L587--611]`) to reduce fragmentation. Bold titled paragraphs ("An empirical property taxonomy," "The simplicity principle," "Relationship to prior work," "Practical implications") have been converted to flowing prose with topic sentences that integrate the content more naturally. Specifically: the taxonomy opening at `[L598]` now uses a topic sentence rather than a bold header; the property categories at `[L600]` flow as continuous prose rather than a bulleted list with italic headers; the prior work transition at `[L609]` flows naturally from the preceding text; and the practical implications at `[L611]` open with a transitional phrase. The Related Work section (Section 2, `[L164--189]`) has similarly been converted from bold-titled blocks to flowing prose with transitional phrases. The section now reads as a cohesive narrative rather than a series of independent blocks.

Specific changes:

- Taxonomy opening rewritten as flowing prose: `[L598]`
- Property categories (solvation, steric, electronic) as flowing text: `[L600]`
- Performance hierarchy rewritten with flowing transition: `[L602]`
- Simplicity principle converted to flowing prose: `[L604]`
- Prior work transition to flowing prose: `[L609]`
- Practical implications transition: `[L611]`
- Related Work bold titles removed, transitional phrases added: `[L164--189]`
- Limitations and Reproducibility headers converted to prose: `[L620, 622]`
- Appendix bold titles converted to prose: `[L1035--1039]`

R3.2: Research questions formulated as actual questions feels too informal. **Response:** We have rephrased RQ1–RQ3 in the Introduction (`[L147--149]`) as declarative objectives rather than interrogative questions:

- RQ1 `[L147]`: "**Property taxonomy**: We characterize which molecular property types benefit from 3D conformer geometry and provide a mechanistic basis for the observed selectivity."

- RQ2 [L148]: "**Representation hierarchy**: We establish how pre-computed conformer ensemble methods compare to 2D fingerprints and end-to-end 3D GNNs (SchNet, PaiNN) across task types."

- RQ3 [L149]: "**Representation complexity**: We test whether complex covariance representations outperform simple summary statistics, grounded in a bias-variance analysis of parameterization complexity versus dataset size."

Specific changes:

- RQ1 rephrased as declarative objective: [L147]
- RQ2 rephrased as declarative objective: [L148]
- RQ3 rephrased as declarative objective: [L149]

R3.3: Performance hierarchy is confused and could benefit from clearer presentation. **Response:** We have substantially rewritten the performance hierarchy presentation in Section 6 at [L602]. The revised text walks through each of the four tiers with explicit evidence and, crucially, explains the progression from tier 4 to tier 1 in terms of how much 3D relational structure is preserved: "DKO discards it into summary statistics, engineered descriptors preserve selected physical invariants, and GNNs retain the full atomic graph." This mechanistic framing—absent from the original—makes the logic of the hierarchy explicit rather than presenting it as an empirical ranking without explanation. The key result—that FP+3D descriptors match SchNet on ESOL (1.000 vs. 1.004)—is highlighted as evidence that carefully engineered features can approach end-to-end learning.

Specific changes:

- Four-tier hierarchy rewritten with mechanistic explanation: [L602]
- Tier progression explained (summary statistics $\rightarrow$ physical invariants $\rightarrow$ full atomic graph): [L602]
- Key FP+3D vs. SchNet comparison highlighted: [L602]
- Abstract hierarchy description updated: [L82]
- Contribution 3 hierarchy description updated: [L157]
- Conclusion hierarchy summary: [L618]

R3.4: Some text too direct; expand into more technical discourse. **Response:** We have expanded terse statements throughout the manuscript into more technical language. Key locations:

- Abstract performance hierarchy: [L82]—now specifies "continuous-filter and equivariant message passing" and "pre-computed feature bottleneck"
- Discussion bias-variance analysis: [L605]—explicit parameter-count calculation ($\sim$180 vs. $\sim$7 samples per parameter), connection to model selection theory
- Hierarchy explanation: [L602]—mechanistic framing of tier progression with evidence citations
- Practical decision framework: [L611]—expanded with specific thresholds and conditions

- Conclusion: `[L618--620]`—tightened with precise statistical claims and explicit practical recommendations
- GNN baseline justification: `[L453]`—technical comparison of continuous-filter vs. equivariant paradigms
- Truncation concern: `[L508]`—technical argument about curse of dimensionality vs. information loss

---

Summary of changes relevant to Reviewer u34v:

- Removed all bold/italic titled paragraphs; converted to flowing prose throughout `[L164--189]`, `[L598--611]`, `[L620--622]`, `[L1035--1039]`
- Rephrased research questions as declarative objectives `[L147--149]`
- Expanded performance hierarchy with mechanistic explanation `[L602]`
- Expanded technical discourse throughout `[L82]`, `[L453]`, `[L508]`, `[L605]`
- All changes marked in blue text